In the Name of God

Deep Generative Models (25732-1)

Homework 1

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Department of Electrical Engineering

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Problem 1 (20 pts). $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ means $\mathbb{E}[e^{\boldsymbol{t}^\top \boldsymbol{x}}] = \exp(\boldsymbol{t}^\top \boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}^\top \boldsymbol{\Sigma} \boldsymbol{t})$ for all $\boldsymbol{t}$ (for $\boldsymbol{\Sigma} \succ 0$ this gives the usual pdf). The mgf determines the law, and jointly Gaussian vectors are independent iff uncorrelated. $h(p) = -\int p \log p$ is the differential entropy.

1.1 (3 pts) Using the mgf, find the law of $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$.

(a) $\boldsymbol{x}, \boldsymbol{y}$ are jointly Gaussian with cross-covariance $\boldsymbol{C} = \mathbb{E}[(\boldsymbol{x} - \boldsymbol{\mu}_x)(\boldsymbol{y} - \boldsymbol{\mu}_y)^\top]$. Find the law of $a\boldsymbol{x} + b\boldsymbol{y}$ and when its covariance is singular (look at $\boldsymbol{x} \perp\!\!\!\perp \boldsymbol{y}$ and $\boldsymbol{y} = \boldsymbol{x}$). For $d = 1$ write $\mathrm{Var}[z]$ with the correlation ρ; what bound on ρ follows and when is it attained?

(b) $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$, $\boldsymbol{A} \in \mathbb{R}^{m \times d}$ with $m > d$ and full column rank. Does $\boldsymbol{A}\boldsymbol{x}$ have a pdf on $\mathbb{R}^m$? With the thin QR factorization $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ describe $\boldsymbol{A}\boldsymbol{x}$ through a nondegenerate Gaussian in $\mathbb{R}^d$, and compute $\mathbb{E}[\|\boldsymbol{A}\boldsymbol{x}\|^2]$ in two ways.

1.2 (4 pts) For symmetric positive definite $\boldsymbol{A}, \boldsymbol{B}$: $\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{a}, \boldsymbol{A})\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{b}, \boldsymbol{B}) = Z\,\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{c}, \boldsymbol{C})$. Find $\boldsymbol{c}, \boldsymbol{C}, Z$; also write $\boldsymbol{C}, \boldsymbol{c}$ with only $(\boldsymbol{A} + \boldsymbol{B})^{-1}$, show the forms agree, and write Z as a Gaussian pdf.

(a) $\theta \sim \mathcal{N}(m_0, s_0^2)$ and, given θ, $x_1, \ldots, x_n$ iid $\mathcal{N}(\theta, s^2)$. Find the posterior of θ, the law of $(x_1, \ldots, x_n)$, the MLE and the MAP (as a mix of m_0 and the MLE), and the limits $n \to \infty$, $s_0 \to \infty$.

1.3 (4 pts) $(\boldsymbol{x}, \boldsymbol{y})$ is jointly Gaussian, $\boldsymbol{\Sigma}_{xx} \succ 0$. Find the law of $\boldsymbol{y}$ given $\boldsymbol{x} = \boldsymbol{x}_0$ without block inversion or completing squares (look at $\boldsymbol{y} - \boldsymbol{\mu}_y - \boldsymbol{K}(\boldsymbol{x} - \boldsymbol{\mu}_x)$ for a suitable $\boldsymbol{K}$). Does the conditional covariance depend on $\boldsymbol{x}_0$? Compare it with $\boldsymbol{\Sigma}_{yy}$ and say when they are equal.

(a) $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{y} \mid \boldsymbol{x} \sim \mathcal{N}(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}, \boldsymbol{S})$, $\boldsymbol{S} \succ 0$. Find the law of $(\boldsymbol{x}, \boldsymbol{y})$, of $\boldsymbol{y}$, and the posterior of $\boldsymbol{x}$ given $\boldsymbol{y}$, with covariance and mean each in two forms: one containing $(\boldsymbol{\Sigma}^{-1} + \boldsymbol{A}^\top \boldsymbol{S}^{-1} \boldsymbol{A})^{-1}$ and one inverting only $\boldsymbol{S} + \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top$. Prove the forms agree and explain the role of the matrix multiplying $\boldsymbol{y} - (\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b})$.

1.4 (5 pts) For $p = \sum_{k \le K} \pi_k \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ find $\mathbb{E}[\boldsymbol{x}]$, $\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^\top]$, $\mathrm{Cov}[\boldsymbol{x}]$ (it splits into a within and a between part).

(a) $d = 1$, $p = \frac{1}{2}\mathcal{N}(-a, 1) + \frac{1}{2}\mathcal{N}(a, 1)$, $a > 0$: find the mean, variance and excess kurtosis $\mathbb{E}[x^4]/\mathrm{Var}[x]^2 - 3$. Is p Gaussian? For which a does p have two modes?

(b) (i) Find $\mathrm{KL}(\mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \| \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1))$. (ii) For any pdf p with mean $\boldsymbol{\mu}_p$, covariance $\boldsymbol{\Sigma}_p \succ 0$ and $g = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, find and prove an exact relation between $\mathrm{KL}(p\|q)$, $\mathrm{KL}(p\|g)$, $\mathrm{KL}(g\|q)$ for Gaussian q. Deduce the best Gaussian q for $\min_q \mathrm{KL}(p\|q)$, apply it to (b), express the minimum via $h(\cdot)$, and conclude a max-entropy property of the Gaussian.

1.5 (4 pts) $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} \succ 0$. Find the law of x_d given $\boldsymbol{x}_{<d}$ in terms of blocks of $\boldsymbol{\Sigma}$, so $p(\boldsymbol{x}) = \prod_d \mathcal{N}(x_d \mid \boldsymbol{w}_d^\top \boldsymbol{x}_{<d}, \sigma_d^2)$. Is this unique? Relate σ_d^2 to determinants of leading blocks of $\boldsymbol{\Sigma}$ and $\det \boldsymbol{\Sigma}$ to the σ_d^2; count the free parameters. Conversely, for any strictly lower-triangular $\boldsymbol{W}$ and $\sigma_d^2 > 0$, find the law of $\boldsymbol{x}$ generated by $x_d = \boldsymbol{w}_d^\top \boldsymbol{x}_{<d} + \varepsilon_d$, $\varepsilon_d \sim \mathcal{N}(0, \sigma_d^2)$ independent.

(a) $|\rho| < 1$, $\boldsymbol{\Sigma}_{ij} = \frac{\sigma^2}{1-\rho^2}\rho^{|i-j|}$. Find $(\boldsymbol{w}_d, \sigma_d^2)$ directly from this formula, and which process has this covariance.

(b) (Hub and leaves) $x_1 \sim \mathcal{N}(0,1)$, $x_j = \beta x_1 + \varepsilon_j$ for $j = 2..D$, $\varepsilon_j \overset{iid}{\sim} \mathcal{N}(0, s^2)$ independent of x_1 ($\beta \neq 0$). Find $\boldsymbol{\Sigma}$, and $\boldsymbol{W}, \boldsymbol{\Delta} = \mathrm{diag}(\sigma_d^2)$ for the orderings $(1, \dots, D)$ and $(D, \dots, 1)$; for the second give weights and variance of the variable in position $m+1$ (a leaf, $0 \le m \le D-2$) and of the hub. Compare the numbers of nonzero weights and $\prod_d \sigma_d^2$, and decide with proof whether a model with the sparsity of the first ordering can represent this distribution in the second.

(c) Let $x_1 \sim \mathcal{N}(0,1)$ and $x_2 \mid x_1 \sim \mathcal{N}(0, e^{ux_1})$, $u \neq 0$. Is (x_1, x_2) jointly Gaussian?

Problem 2 (15 pts). Let $x_1, \dots, x_N \overset{iid}{\sim} p$ and $f_N = \frac{1}{N}\sum_n f(x_n)$, the Monte Carlo (MC) estimate of $\mathbb{E}_p[f]$.

2.1 (5 pts) Is f_N unbiased? Find its variance and, with Chebyshev, the weak law with its rate.

(a) $X \sim \mathcal{N}(1, 4)$ (variance 4), $f(x) = (x-1)^2$. Find $\mathbb{E}[f]$, $\mathrm{Var}[f]$ and the law of f_N (a scaled χ^2_N). For $\Pr(|f_N - 4| \ge \frac{1}{2}) \le 0.05$ find a sufficient N from (i) Chebyshev, (ii) the CLT (use 1.96), (iii) a Chernoff bound derived from the mgf of χ^2_N for both tails (exponent in N and the relative deviation t; value to the nearest 50). Which are guaranteed, and why do they differ?

(b) (Importance sampling.) $p = \mathcal{N}(0,1)$, $f(x) = x^2$, $x_n \overset{iid}{\sim} q = \mathcal{N}(0, s^2)$, $\hat{I}_N = \frac{1}{N}\sum_n f(x_n)\frac{p(x_n)}{q(x_n)}$. Is it unbiased? Find $\mathrm{Var}[\hat{I}_N]$ (for which s is it finite?), the best s and the minimal value, and the optimal proposal $q^\star$ over all densities with its variance.

2.2 (4 pts) Let $\ell(x) = -\ln p_\theta(x)$, $\widehat{\mathrm{PPL}}_N = \exp(\frac{1}{N}\sum_{n \le N} \ell(x_n))$ for $x_n \overset{iid}{\sim} p_{\mathrm{data}}$, and $\mathrm{PPL} = \exp(\mathbb{E}[\ell])$. Is $\widehat{\mathrm{PPL}}_N$ unbiased for PPL? Compare $\mathbb{E}[\widehat{\mathrm{PPL}}_N]$ with PPL for every N, say when they are equal, and prove consistency.

(a) Find c in $\mathbb{E}[\widehat{\mathrm{PPL}}_N] = \mathrm{PPL}(1 + c/N + o(1/N))$ in terms of $\mathrm{Var}[\ell]$, with the moment assumption.

(b) $p_{\mathrm{data}} = \mathcal{N}(0, s^2)$, $p_\theta = \mathcal{N}(0, \sigma^2)$, $r = s^2/\sigma^2$: find $\mathbb{E}[\widehat{\mathrm{PPL}}_N]$ in closed form, for which N it is finite, check it against (a), (b), and comment on $r > 1$.

2.3 (3 pts) For discrete p, q let $m = \frac{p+q}{2}$, $\mathrm{JS}(p \,\|\, q) = \frac{1}{2}\mathrm{KL}(p \,\|\, m) + \frac{1}{2}\mathrm{KL}(q \,\|\, m)$. Write $\mathrm{JS}(p \,\|\, q)$ with entropies and as a mutual information of a suitable pair (Z, X); find its range (natural log) and when the extremes occur.

(a) Compare p, q with m pointwise for the upper bound again. Is $\mathrm{JS}(p \,\|\, q)$ finite when $\mathrm{KL}(p \,\|\, q) = \infty$? Give an example where $\mathrm{KL}(p \,\|\, q) = \infty$ and $\mathrm{JS}(p \,\|\, q)$ hits its maximum. For which log base is the often quoted bound $0 \le \mathrm{JS}(p \,\|\, q) \le 1$ tight, and what is the supremum in nats?

2.4 (3 pts) For $a \in (0,1)$ study $\mathrm{KL}(\mathrm{Ber}(a) \,\|\, \mathrm{Ber}(c))$ as $c \to 0^+$ (leading term and the constant after it) and $\mathrm{KL}(\mathrm{Ber}(c) \,\|\, \mathrm{Ber}(a))$ (its limit). What does it say about symmetry?

(a) Does KL satisfy the triangle inequality? For $1 > a > b > 0$ prove what happens to $\mathrm{KL}(\mathrm{Ber}(a) \,\|\, \mathrm{Ber}(c)) \le \mathrm{KL}(\mathrm{Ber}(a) \,\|\, \mathrm{Ber}(b)) + \mathrm{KL}(\mathrm{Ber}(b) \,\|\, \mathrm{Ber}(c))$ as $c \to 0^+$, and give a numerical triple.

Problem 3 (15 pts). Let $X = (X_1, \dots, X_n)$, each $X_i \in \{1, \dots, K\}$. The number of free parameters of a family of pmfs is its dimension. Write $c_i = x_{\max(1,i-m):i-1}$ for the window of the m variables before x_i.

3.1 (4 pts) $\mathcal{Q}_m^{\mathrm{nh}}$ is the family $q(x) = \prod_i q_i(x_i \mid c_i)$ with a separate table q_i for each i ($0 \le m < n$). $\mathcal{Q}_m^{\mathrm{h}} \subset \mathcal{Q}_m^{\mathrm{nh}}$ has $q_i = T$ for all $i > m$ and an arbitrary pmf for $x_1, \dots, x_m$. Find the numbers N_m^{nh}, N_m^{h} of free parameters (check $m = 0$ and $m = n - 1$), their difference, and interpret $n = m + 1$.

(a) ($K = 2$) A mixture transition model has $\Pr(x_i{=}1 \mid x_{i-1}{=}a, x_{i-2}{=}b) = c_{ab} = \lambda t_1(a) + (1 - \lambda) t_2(b)$, $\lambda \in [0, 1]$, $t_1, t_2 : \{0, 1\} \to [0, 1]$ (5 parameters; a free table has 4). Find exactly which $(c_{00}, c_{01}, c_{10}, c_{11})$ it can produce and the dimension of that set. What does it say about the 5 parameters?

3.2 (5 pts) Let $p > 0$ be a pmf on $\{1, \dots, K\}^n$ and $q^\star = \prod_i p(x_i \mid c_i)$ (conditionals of p itself). Find and prove an exact relation between $\mathrm{KL}(p\|q)$, $\mathrm{KL}(p\|q^\star)$ and the tables q_i for every $q \in \mathcal{Q}_m^{\mathrm{nh}}$. Which q is closest to p; is it unique?

(a) Write $\mathrm{KL}(p\|q^\star)$ with conditional mutual informations $I(X; Y \mid Z) = \mathbb{E}[\log \frac{p(x,y|z)}{p(x|z)p(y|z)}]$ (variables, index range). When is it zero? What is it for $m = 0$?

(b) From N iid sequences derive the maximum likelihood tables in $\mathcal{Q}_m^{\mathrm{nh}}$ (Lagrange multipliers) and relate them to (a).

(c) In $\mathcal{Q}_m^{\mathrm{h}}$ find the KL-closest member to p (initial pmf and $T^\star$ via $p_i(c, k) = \Pr_p(c_i = c, x_i = k)$), write $T^\star$ as an average of the $p_i(k \mid c)$ with its weights, give the extra KL over (a) as a sum of KL terms, and say when it vanishes.

3.3 (6 pts) Figure 1 shows a model on seven binary variables, $p(\boldsymbol{x}) = \prod_v p(x_v \mid \boldsymbol{x}_{\mathrm{pa}(v)})$, with $\mathrm{pa}(v)$ the variables having an arrow into v. Read off the factorization, count the free parameters, and prove

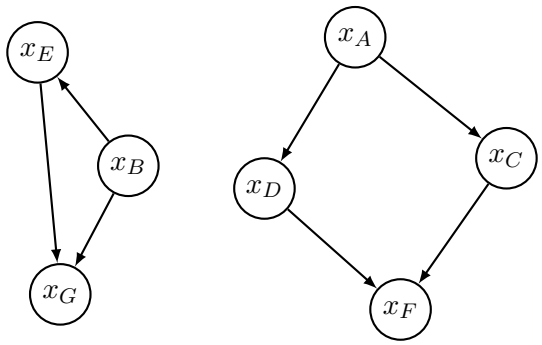


Figure 1: An arrow $u \to v$ means $u \in \mathrm{pa}(v)$.

that for any acyclic graph $\prod_v p(x_v \mid \boldsymbol{x}_{\mathrm{pa}(v)})$ is normalized.

(a) An ordering is valid if every variable comes after its parents. Count the valid orderings and find the probability that a random ordering of the 7 variables is valid.

(b) (Rooted forest: every node has at most one parent.) Find and prove a formula for the number of valid orderings in terms of n and the subtree sizes h_v. Apply it to the forest from the figure without $C \to F$ and $E \to G$. Then, from this count only, find the fraction of its orderings that survive when $E \to G$ is put back, when $C \to F$ is put back, and when both are, and compare with (b).

(c) Two binary variables get $q(x_1, x_2) = p(x_1 \mid x_2)r(x_2 \mid x_1)$, $p(x_1{=}1 \mid x_2{=}b) = \theta_b$, $r(x_2{=}1 \mid x_1{=}a) = \varphi_a$. Compute $Z = \sum q$. When is $Z = 1$? Why does $Z \neq 1$ break maximum likelihood?

Problem 4 (20 pts). $\boldsymbol{x} \in \{0,1\}^D$, $p(\boldsymbol{x}) = \prod_d p(x_d \mid \boldsymbol{x}_{<d})$ with Bernoulli conditionals.

FVSBN (fully visible sigmoid belief network): $p(x_1) = \mathrm{Ber}(\sigma(b_1))$, $p(x_d \mid \boldsymbol{x}_{<d}) = \mathrm{Ber}(\sigma(b_d + \boldsymbol{w}_d^\top \boldsymbol{x}_{<d}))$.

NADE (neural autoregressive density estimator): $\boldsymbol{a}_d = \boldsymbol{c} + \sum_{k<d} \boldsymbol{W}_{:,k} x_k$, $\boldsymbol{h}_d = \phi(\boldsymbol{a}_d)$, $p(x_d \mid \boldsymbol{x}_{<d}) = \mathrm{Ber}(\sigma(\boldsymbol{v}_d^\top \boldsymbol{h}_d + b_d))$ for $d \geq 2$, $p(x_1) = \mathrm{Ber}(\sigma(b_1))$. Here $\boldsymbol{W} \in \mathbb{R}^{D_h \times (D-1)}$ (columns shared by all conditionals), $\boldsymbol{c}, \boldsymbol{v}_d \in \mathbb{R}^{D_h}$, ϕ is a differentiable elementwise nonlinearity.

4.1 (4 pts) Find the number of parameters of FVSBN and NADE. For which D_h is NADE smaller? Largest such D_h for $D = 784$?

(a) Count scalar multiplications for $\log p(\boldsymbol{x})$ in FVSBN, in NADE recomputing each $\boldsymbol{a}_d$, and in NADE with the recursive update $\boldsymbol{a}_{d+1} = \boldsymbol{a}_d + \boldsymbol{W}_{:,d} x_d$; evaluate for $D = 784$, $D_h = 500$.

(b) Why does sampling still need D sequential stages although it can use the same recursion, while density evaluation does not?

4.2 (4 pts) For one $\boldsymbol{x}$ let $L = -\log p_\theta(\boldsymbol{x})$ (NADE), $z_d = \boldsymbol{v}_d^\top \boldsymbol{h}_d + b_d$ ($z_1 = b_1$), $\delta_d = \sigma(z_d) - x_d$. Derive the gradients of L w.r.t. $b_d, \boldsymbol{v}_d, \boldsymbol{h}_d, \boldsymbol{a}_d, \boldsymbol{c}, \boldsymbol{W}_{:,k}$ (watch which conditionals each parameter affects).

(a) How can all $\partial L / \partial \boldsymbol{W}_{:,k}$ be computed at the cost of one forward pass (which order, which idea)? With N samples and the prior $\theta \sim \mathcal{N}(\boldsymbol{0}, \tau^2 \boldsymbol{I})$ find the MAP objective and weight decay. For $\phi = \tanh$ give a lower bound (in D_h) on the number of parameter vectors defining the same p_θ by architectural symmetries, and prove the invariances.

4.3 (5 pts) Let $\mathcal{P}_D$ be the pmfs on $\{0,1\}^D$ with positive entries (dimension $2^D - 1$). For a fixed ordering, $\ell_d(\boldsymbol{x}_{<d}) = \log \frac{p(x_d=1 \mid \boldsymbol{x}_{<d})}{p(x_d=0 \mid \boldsymbol{x}_{<d})}$. Characterize the set $\mathcal{F}_D$ of pmfs a FVSBN can represent through the ℓ_d, decide whether parameters $\to$ pmf is injective, find $\dim \mathcal{F}_D$ and for which D it equals $\mathcal{P}_D$.

(a) $\theta = 0.1$, $p(\boldsymbol{x}) = \theta/4$ if $x_1 \oplus x_2 \oplus x_3 = 1$ and $(1-\theta)/4$ otherwise ($\oplus$: sum mod 2). Can some FVSBN represent p for some ordering? Prove it; compute the second-order interaction of the last variable's log-odds.

(b) D bits: $x_1, \ldots, x_{D-1}$ iid uniform and x_D such that the total parity is odd with probability θ. Can a NADE with $\phi = \mathrm{ReLU} = \max(0, \cdot)$ and $D_h = D - 1$ represent it exactly? If so give $\boldsymbol{c}, \boldsymbol{W}, \boldsymbol{v}_d, b_d$ and check them.

4.4 (7 pts) MADE (masked autoencoder for distribution estimation) masks a feed-forward net: $\boldsymbol{h} = \phi(\boldsymbol{c} + (\boldsymbol{M}^W \odot \boldsymbol{W})\boldsymbol{x})$, $\boldsymbol{\lambda} = \sigma(\boldsymbol{b} + (\boldsymbol{M}^V \odot \boldsymbol{V})\boldsymbol{h})$, $p(\boldsymbol{x}) = \prod_d \mathrm{Ber}(x_d \mid \lambda_d)$. Input i has degree $m^0(i) \in \{1..D\}$ (its position in the ordering), output d has degree $m^0(d)$, hidden unit k has $m^1(k) \in \{1..D-1\}$, and

$$[\boldsymbol{M}^W]_{ki} = \mathbf{1}\{m^1(k) \geq m^0(i)\}, \qquad [\boldsymbol{M}^V]_{dk} = \mathbf{1}\{m^0(d) > m^1(k)\}.$$

Figure 2 gives the degrees for $D = 4$, $H = 6$. Write $\boldsymbol{M}^W \in \{0,1\}^{6\times 4}$, $\boldsymbol{M}^V \in \{0,1\}^{4\times 6}$ for the figure and count their nonzeros. Which earlier variables can each $p(x_d \mid \cdot)$ really depend on? Is the network as expressive as a general AR model for its ordering? If not, what is lost and why; for general D and one hidden layer find (with proof) the smallest H that loses nothing the ordering allows.

(a) For any depth and any hidden degrees (same rule between hidden layers as at the input), prove which inputs λ_d can depend on and that $p(\boldsymbol{x})$ is normalized. What fails in the figure if the output rule is made non-strict, $\mathbf{1}\{m^0(d) \geq m^1(k)\}$?

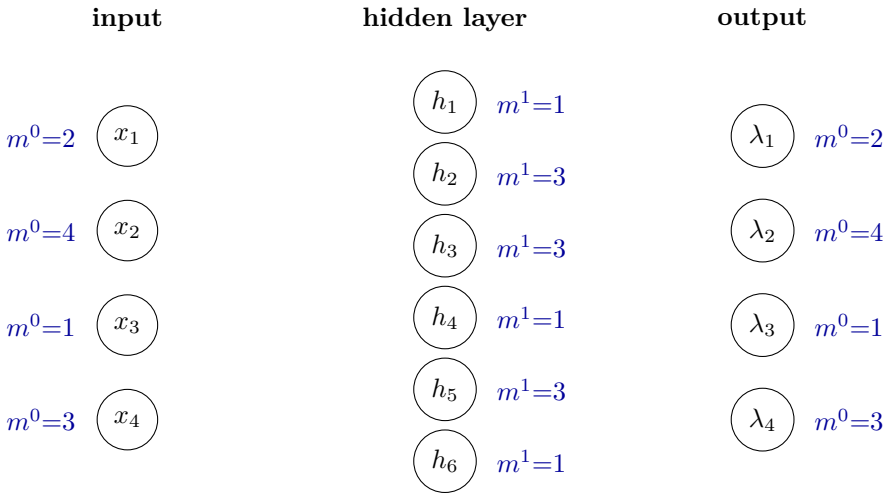


Figure 2: Degrees (connections follow from the masks).

(b) Let the H hidden degrees be iid uniform on $\{1..D-1\}$. Find the expected nonzeros of $\boldsymbol{M}^W, \boldsymbol{M}^V$; the probability that an allowed dependence (degree $r \to j > r$) is connected; the probability that all are; the expected H needed until all are connected, and its value for $D = 784$.

Problem 5 (15 pts). Real NADE is the real-valued NADE of Problem 4. With $\boldsymbol{a}_d = \boldsymbol{c} + \sum_{k<d} \boldsymbol{W}_{:,k} x_k$, $\boldsymbol{h}_d = \phi(\boldsymbol{a}_d)$:

$$p(x_d \mid \boldsymbol{x}_{<d}) = \mathcal{N}(x_d \mid \mu_d, e^{s_d}), \quad \mu_d = \boldsymbol{v}_d^\top \boldsymbol{h}_d + b_d,\ s_d = \boldsymbol{u}_d^\top \boldsymbol{h}_d + e_d\ (d \geq 2), \quad \mu_1 = b_1,\ s_1 = e_1.$$

Note that e^{s_d} is the variance. Let $\ell(x; \mu, s) = -\log \mathcal{N}(x \mid \mu, e^s)$.

5.1 (3 pts) Find $\partial\ell/\partial\mu$, $\partial\ell/\partial s$ (call them δ^μ, δ^s) and derive the gradients of $L(\boldsymbol{x}) = -\log p(\boldsymbol{x})$ w.r.t. $b_d, \boldsymbol{v}_d, e_d, \boldsymbol{u}_d, \boldsymbol{h}_d, \boldsymbol{a}_d, \boldsymbol{c}, \boldsymbol{W}_{:,k}$.

(a) A constant model (μ, s) is fitted to iid scalars $x_n \sim \mathcal{N}(\mu_0, \sigma_0^2)$, $n \leq N$. Find the stationary points of $\sum_n \ell(x_n; \mu, s)$, decide if it is the global minimum, and compute $\mathbb{E}[e^{\hat{s}}]$ there.

5.2 (4 pts) Find the Hessian of ℓ in (μ, s) and its determinant. What does it say about convexity?

(a) For $\eta_1 = \mu e^{-s}$, $\eta_2 = -\frac{1}{2} e^{-s}$ is ℓ convex in η? Prove it (use $A(\eta)$ in $\frac{1}{\sqrt{2\pi}} e^{\eta_1 x + \eta_2 x^2 - A(\eta)}$; find it).

(b) In which senses is ℓ convex in μ and in s? What follows for the NLL in $(\boldsymbol{v}_d, b_d)$ with $(\boldsymbol{u}_d, e_d, \boldsymbol{h}_d)$ fixed and in $(\boldsymbol{u}_d, e_d)$ with the rest fixed? Is the NLL jointly convex in $(\boldsymbol{v}_d, \boldsymbol{u}_d)$? Prove it or give a counterexample.

5.3 (4 pts) $D = 1$: is the NLL bounded below with one training point? With $N \geq 2$ distinct points (best bound in N and $S = \sum (x_n - \bar{x})^2$)?

(a) $D = 2$, distinct first coordinates, hidden layer wide enough to interpolate $\mu_2(\cdot), s_2(\cdot)$ at the N points. Is the training NLL bounded below? Prove it. Why can the average NLL of discrete data not go below the entropy $H(p_{\text{data}})$ while continuous data has no such floor?

(b) Does a variance floor $s_d \geq s_{\min}$ change (b)? Find a lower bound on $-\log p_\theta(\boldsymbol{x})$.

5.4 (4 pts) Use a mixture $\sum_{c \leq C} \pi_c \mathcal{N}(x_d \mid \mu_c, e^{s_c})$, $\pi = \text{softmax}(\ell_1..\ell_C)$, with heads $\ell_c = \boldsymbol{a}_c^\top \boldsymbol{h}_d + \alpha_c$, $\mu_c = \boldsymbol{v}_c^\top \boldsymbol{h}_d + b_c$, $s_c = \boldsymbol{u}_c^\top \boldsymbol{h}_d + e_c$. Let $L(x)$ be the NLL of one conditional and $\gamma_c = \pi_c \mathcal{N}(x \mid \mu_c, e^{s_c}) / \sum_j \pi_j \mathcal{N}(x \mid \mu_j, e^{s_j})$ the responsibility of component c.

(a) Derive $\partial L/\partial \ell_c$, $\partial L/\partial \mu_c$, $\partial L/\partial s_c$, $\partial L/\partial \boldsymbol{h}$ with γ_c; compare with 5.1 and find a linear relation among the $\partial L/\partial \ell_c$.

(b) Is the likelihood bounded above for $C \geq 2$, $D = 1$, $N \geq 2$ distinct points? Prove it.

(c) Is the NLL convex? Decide with an example ($C = 2$, one observation $x = 0$, $\pi = (\frac{1}{2}, \frac{1}{2})$, $s_1 = s_2 = 0$, free means), and count the parameter vectors giving the same distribution by relabeling components.

Problem 6 (15 pts). A generative classifier models $p(y)p(\boldsymbol{x} \mid y)$ and uses Bayes' rule; a discriminative one (logistic regression) models $p(y{=}1 \mid \boldsymbol{x}) = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$ directly. Let $y \in \{0, 1\}$, $p(y{=}1) = \pi$, $\boldsymbol{x} \mid y{=}c \sim \mathcal{N}(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ in $\mathbb{R}^D$, $\boldsymbol{\Delta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0$, $\bar{\boldsymbol{\mu}} = \frac{1}{2}(\boldsymbol{\mu}_0 + \boldsymbol{\mu}_1)$.

6.1 (4 pts) Find a in $p(y{=}1 \mid \boldsymbol{x}) = \sigma(a(\boldsymbol{x}))$. For $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}$ show a is affine and find $\boldsymbol{w}, b$; for $\boldsymbol{\Sigma}_0 \neq \boldsymbol{\Sigma}_1$ find the general (quadratic) form.

(a) Count the parameters of the shared-covariance generative model and of logistic regression, and the dimension of the set of generative parameters giving the same $(\boldsymbol{w}, b)$.

6.2 (5 pts) Let $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}$; only the coordinates in V are observed, those in M are masked; $\boldsymbol{S}_{V|M} = \boldsymbol{\Sigma}_{VV} - \boldsymbol{\Sigma}_{VM}\boldsymbol{\Sigma}_{MM}^{-1}\boldsymbol{\Sigma}_{MV}$. Find $p(y{=}1 \mid \boldsymbol{x}_V)$ under the generative model.

(a) A classifier $\sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$ trained on full data is applied after replacing the masked coordinates by $\bar{\boldsymbol{\mu}}_M$. Compare the coefficient it applies to $\boldsymbol{x}_V$ with the Bayes-optimal one of (a): express the difference with $\boldsymbol{S}_{V|M}$ and blocks of $\boldsymbol{\Sigma}, \boldsymbol{\Delta}$, find when they agree for all $\boldsymbol{x}_V$, and interpret.

6.3 (6 pts) Data $(\boldsymbol{x}_n, y_n)$, $n \leq N$, with N_c points in class c. Derive the MLE of $\pi, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1, \boldsymbol{\Sigma}$ for the shared-covariance model (assume the pooled scatter $\hat{\boldsymbol{\Sigma}}$ is positive definite). For $\boldsymbol{\Sigma}$ avoid matrix-derivative tables: compare the objective at $\boldsymbol{\Sigma}$ and $\hat{\boldsymbol{\Sigma}}$ via the eigenvalues of $\boldsymbol{\Sigma}^{-1/2}\hat{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}^{-1/2}$.

(a) For logistic regression find the gradient and Hessian of the NLL; is it convex? The data are separable if some $(\boldsymbol{w}_0, b_0)$ has $\boldsymbol{w}_0^\top \boldsymbol{x}_n + b_0 > 0$ for class 1 and < 0 for class 0. Is there a minimizer for separable data? Prove it, describe the minimizing sequences, and contrast with (a).

(b) With $\frac{\lambda}{2}\|\boldsymbol{w}\|^2$ added ($\lambda > 0$, bias unpenalized), is there a minimizer, and is it unique, even for separable data?